Activity 1 Notes+ Reminders:

Video 1: Ohm's Law
V=IR
- Resistors in Series can be directly added R1+R2+R3.
- Current following to a junction must be equal to the total current that is leaving a junction. Current law.

Video 2: Khan Academy Ohm's Law
- Voltage: electric potential energy per unit charge (Volts)
- Current: charge per unit time (Amperes). Current flows from positive -> negative when drawing circuits although electrons behave differently.

Video 3: Compound Circuits
- Current moving through a resistor heats it up. As charges pass through the resistor, their energy is spread out through the resistor in the form of heat which cause the rise in temperature. The change in electrical potential energy from the start to the end of the resistor corresponds to an increase in thermal energy of the resistor it passes through.
- Power : the amount of energy gained per time. The change in electric potential energy per time. It is the rate at which energy is transferred.
- Power : current squared multiplied by resistance. Or change in voltage squared divided by the resistance.
- Electrical potential energy : charge multiplied by the electric potential q * V.

Video 4: Voltage Dividers
- Resistors in parallel R_total = 1/((1/R1)+(1/R2)+...(1/Rn))
- Two resistors in parallel can be calculated = R1*R2/(R1+R2)
- If you want to halve the value of your resistance, put two resistors in parallel.
- If you want to decrease the value of the resistor by 10% put it in parallel with a resistor x10 the value.
- Output Voltage V_out= R2/(R1+R2) * V_in for a resistor voltage divider.
- Current Dividers (with two resistors in parallel): I1 = R2/(R1+R2) * I
- Voltage Source:
  - Thevenin Voltage = V_open
  - Thevenin Resistance = V_open/I_short
- Current Source:
  - Norton Current: Thevenin Voltage / Thevenin Resistance
  - Norton Resistance: Thevenin Resistance

Video 5: Kirchhoff's Laws
- Kirchhoff's Current Law (KCL): Sum of currents into a junction equals the sum of currents out of the junction. Current_in = Current_out.
- Kirchhoff's Voltage Law (KVL): The sum of applied voltages around a loop equals the sum of the voltage drops around that loop.

- The sum of voltages around a loop equates to 0 (KVL).

Video 6: Intro to DC/DC
- Average KVL, average KCL, conservation of energy and periodic state relations can help me figure out the relationship between the DC input and DC output in a power converter in periodic steady state. This can help us understand a power converter, what is it doing adn what it can and can't do.
- Mistakes to look out for: do not sub in average for instantaneous waveforms.
- Boost converter: can take a small input voltage and boost it to a bigger output voltage.
- Buck converter: lowers voltage.

Summary:

- Use Ohm's Law to calcuate current through a resistor when voltage and resistance are known for a circuit.
- Use power equations to check how much heat a component may need to dissipate.
- Use KVL and KCL to analyze circuits that cannot be reduced to simpler designs/combinations.
- Use a voltage divider to determine a lower voltage from a higher voltage.
- A buck converter reduces a higher DC input voltage into a lower DC output whereas a boost converter does the opposite.